

PDF ! Lean 4 ! L^AT_EX Round-Trip Demo

May 10, 2026

1 Introduction

This document was produced by typesetting a `.tex` source with `pdflatex`, then feeding the resulting PDF (after `mutool clean -d`) into our Lean-4 parser, which emitted a new `.tex` that was again compiled with `pdflatex`. If you can read this paragraph, the round trip succeeded.

2 What the parser handles

Lexical layer: numbers, names, literal strings, hex strings, comments.

All eight PDF object types, plus indirect references.

Cross-reference tables and (uncompressed) cross-reference streams; Prev chains for incremental updates.

Object-stream resolution (`/Type` `/ObjStm`).

Page tree walk and content-stream parsing.

Text-state operators: `BT`, `ET`, `Tf`, `Td`, `TD`, `Tm`, `T*`, `Tj`, `TJ`, `'`, `”`.

Graphics-state nesting (`q/Q`) and `cm`

3 Out of scope (preprocess required)

Flate/LZW decompression — run `mutool clean -d` first.

Encryption (RC4 / AES).

Font CMaps — only the predened ASCII encodings render correctly.

Vector graphics, images, colour.

4 Page Two

A few mathematical expressions to push the parser:

$$\int_0^1 e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} = 2^n:$$

A bullet list:

- alpha

- beta

- gamma

A numbered list:

- one
- two
- three